# GITLAB CI - APPLY AND DESTROY

## What is GitLab CI?

**GitLab CI/CD** is used to automate tasks like:

- Build
- Test
- Deploy
- Infrastructure creation (Terraform)

CI/CD runs pipelines automatically when you push code.

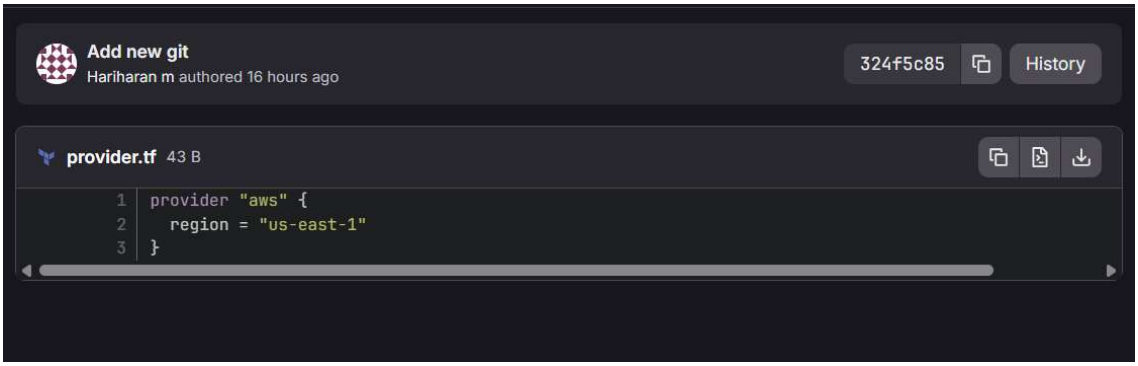
## What is terraform apply?

**Terraform apply = Create or update infrastructure**

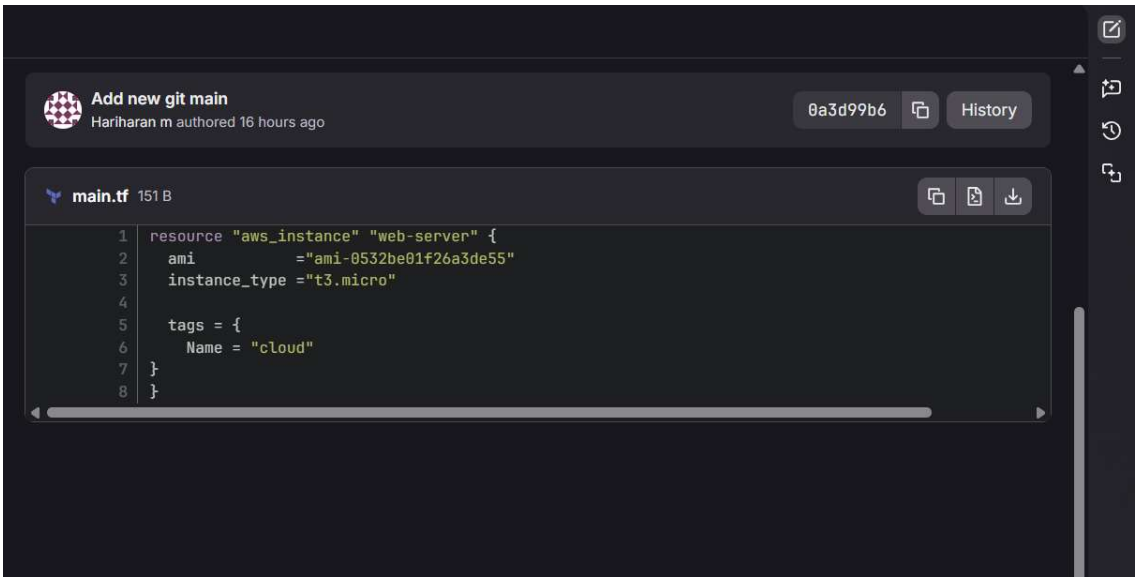
It will:

- Create new resources (EC2, VPC, S3, etc.)
- Modify existing resources
- Keep infra in the desired state

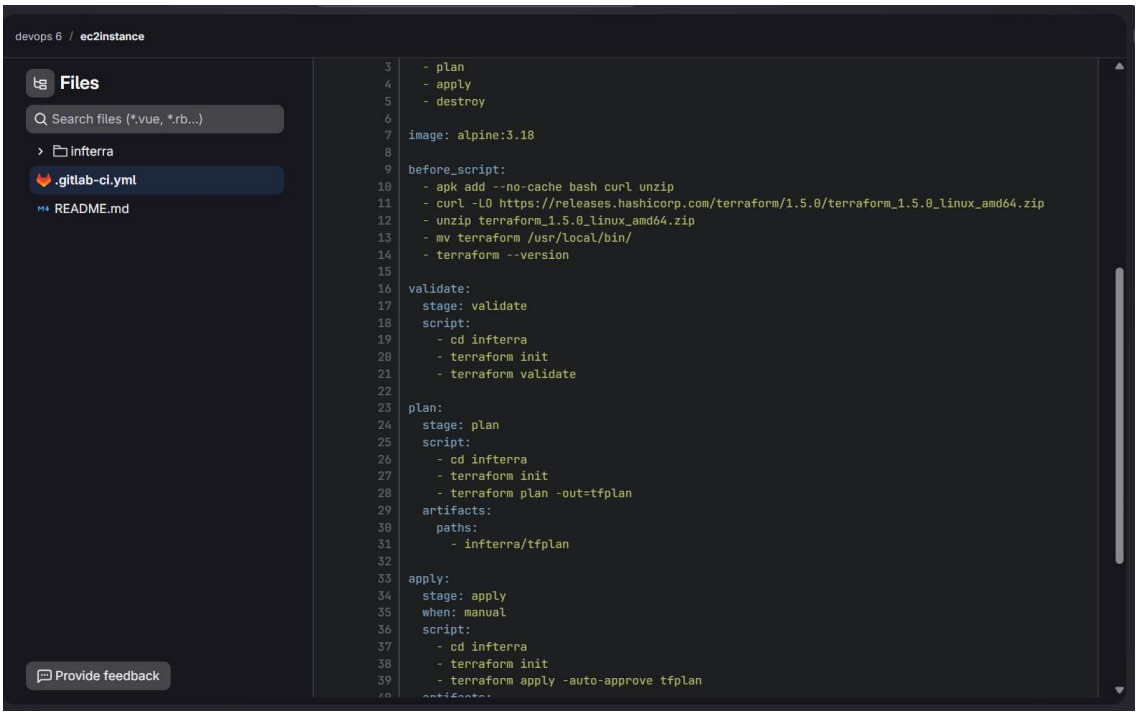
- Using “provider.tf” to create a file in infterra directory . provider aws and region is “us-east-1”.

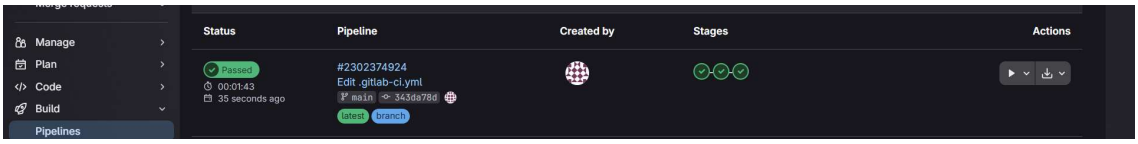


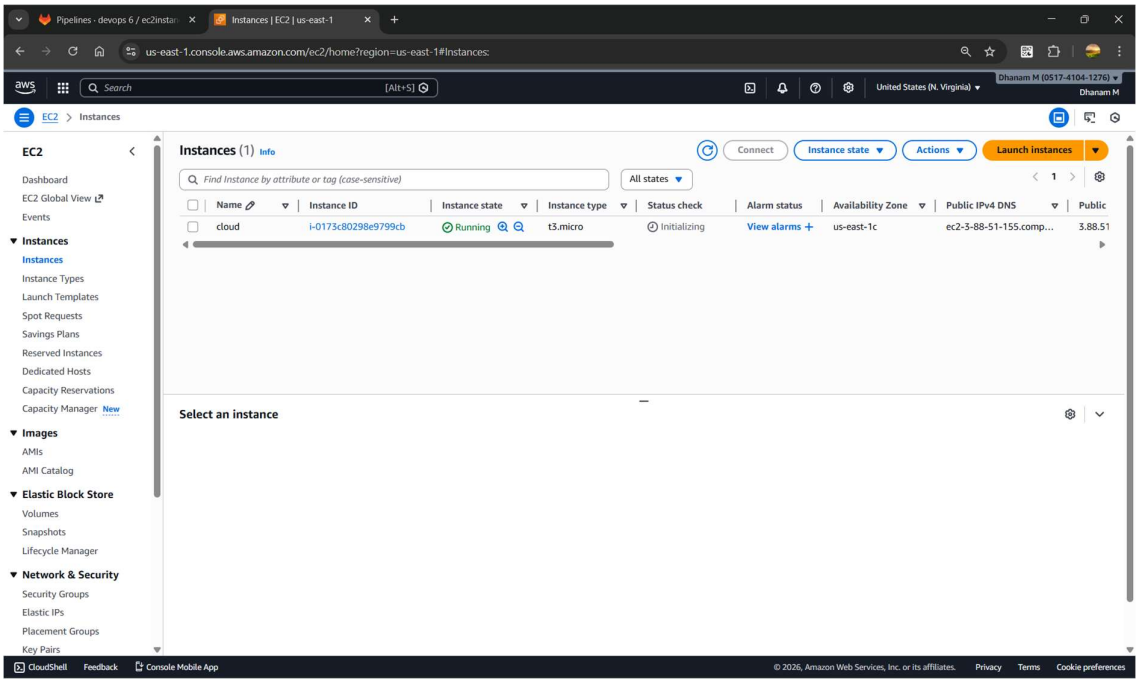

- Using “main.tf” to create a file in infterra directory. To create an ec2 instances in aws using ami id and ami instance type.




- The .gitlab-ci.yml file is a YAML-based configuration file that defines the structure and execution of the Continuous Integration/Continuous Delivery (CI/CD) pipelines in GitLab. It is placed in the root directory of a GitLab repository and tells the GitLab Runner application what tasks to perform, and in what order.





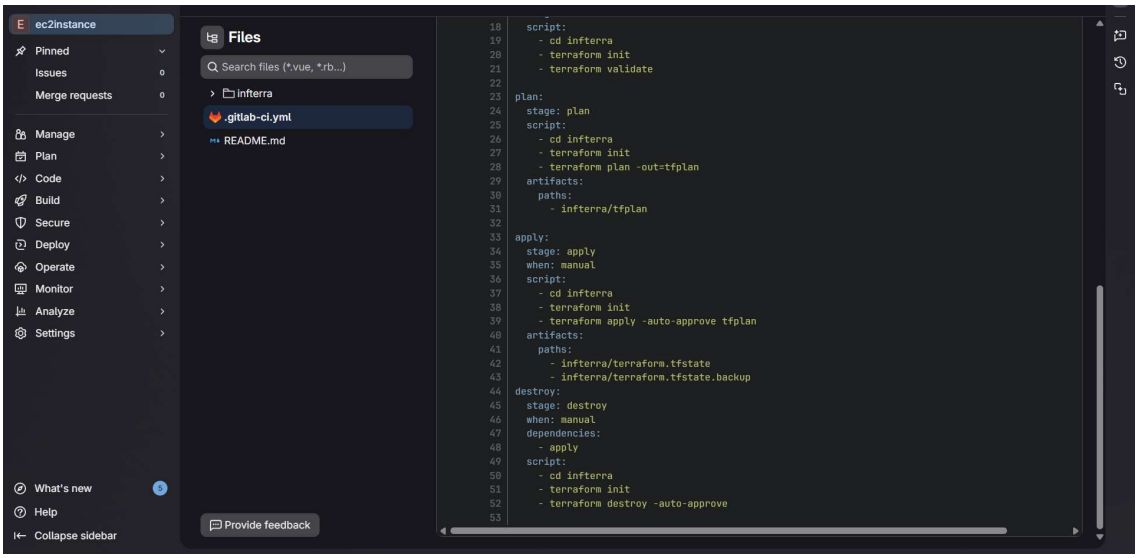



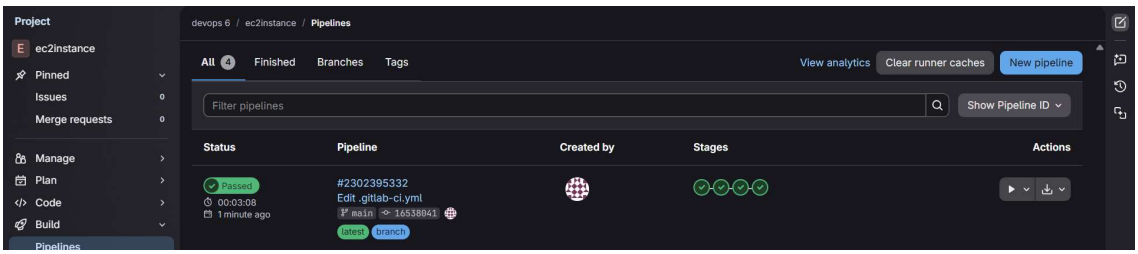

## What is terraform destroy?

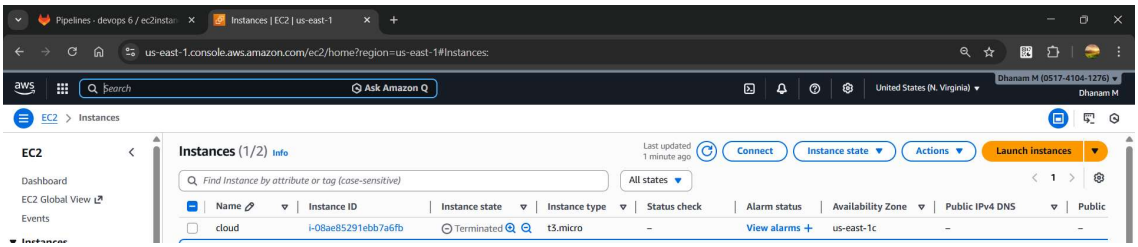
**Terraform destroy = Delete all infrastructure**

It will:

- Remove EC2
- Delete VPC
- Delete load balancers
- Clean up everything created by Terraform


ec2instance
Pinned
Issues 0
Merge requests 0
Manage
Plan
Code
Build
Secure
Deploy
Operate
Monitor
Analyze
Settings
What's new 5
Help
Collapse sidebar
Files
Search files (*.vue, *.rb...)
infterra
.gitlab-ci.yml
README.md
Provide feedback
18 script:
19 - cd infterra
20 - terraform init
21 - terraform validate
22
23 plan:
24 stage: plan
25 script:
26 - cd infterra
27 - terraform init
28 - terraform plan -out=tfplan
29 artifacts:
30 paths:
31 - infterra/tfplan
32
33 apply:
34 stage: apply
35 when: manual
36 script:
37 - cd infterra
38 - terraform init
39 - terraform apply -auto-approve tfplan
40 artifacts:
41 paths:
42 - infterra/terraform.tfstate
43 - infterra/terraform.tfstate.backup
44 destroy:
45 stage: destroy
46 when: manual
47 dependencies:
48 - apply
49 script:
50 - cd infterra
51 - terraform init
52 - terraform destroy -auto-approve
53


Project
ec2instance
Pinned
Issues 0
Merge requests 0
Manage
Plan
Code
Build
Pipelines
devops 6 / ec2instance / Pipelines
All 4
Finished
Branches
Tags
View analytics
Clear runner caches
New pipeline
Filter pipelines
Show Pipeline ID
Status
Pipeline
Created by
Stages
Actions
Passed
00:03:08
1 minute ago
#2302395332
Edit .gitlab-ci.yml
main 16538041
latest
branch


Pipelines · devops 6 / ec2instan
Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
aws
Search
Ask Amazon Q
United States (N. Virginia)
Dhanam M (0517-4104-1276)
Dhanam M
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances (1/2) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Public
cloud
i-08ae85291ebb7a6fb
Terminated
t3.micro
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View alarms +
us-east-1c
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